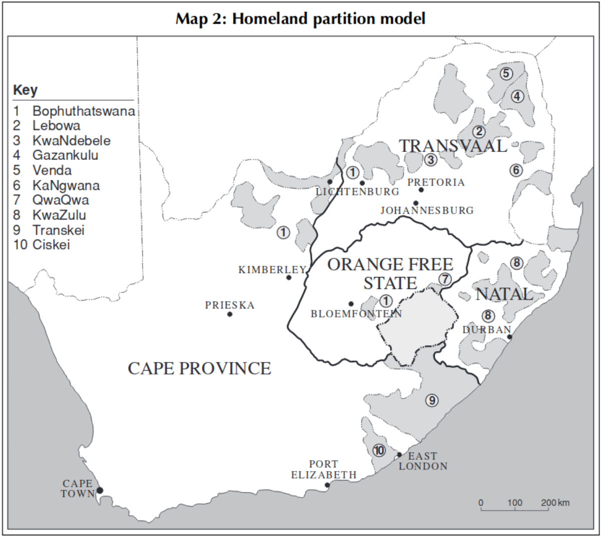
Dinkelman (2011): Data

Covariates in 1996	Means (standard deviation)			Differences in means (standard error)		
					By gradient	
	Full sample (1)	Eskom project (2)	No project (3)	Columns 2–3 (4)	No controls (5)	Controls (6)
Poverty rate	0.61	0.59	0.61	−0.024**	0.00	0.002
	(0.19)	(0.17)	(0.20)	(0.01)	(0.00)	(0.00)
Female-headed HHs	0.55	0.55	0.55	0.00	0.005***	0.001
	(0.13)	(0.12)	(0.13)	(0.01)	(0.00)	(0.00)
Adult sex ratio ($N_{females}/N_{males}$)	1.48	1.41	1.49	−0.080***	0.011***	0.004**
	(0.28)	(0.25)	(0.29)	(0.02)	(0.00)	(0.00)
Indian, white adults × 10	0.00	0.00	0.00	0.00	0.000	0.000
	(0.01)	(0.00)	(0.01)	(0.00)	(0.00)	(0.00)
Kilometers to road	37.95	35.62	38.54	−2.917**	−0.201	−0.156
	(24.57)	(24.18)	(24.64)	(1.44)	(0.41)	(0.18)
Kilometers to town	38.57	36.34	39.13	−2.790***	0.278	0.180
	(18.12)	(15.34)	(18.72)	(1.06)	(0.41)	(0.13)
Men with high school	0.06	0.08	0.06	0.016***	−0.002***	−0.003**
	(0.05)	(0.05)	(0.05)	(0.00)	(0.000)	(0.00)
Women with high school	0.07	0.08	0.06	0.020***	−0.002***	0.000
	(0.05)	(0.05)	(0.05)	(0.00)	(0.000)	(0.00)
Household density	22.05	32.56	19.41	13.152***	−0.523*	−0.944***
	(30.48)	(49.31)	(22.75)	(1.76)	(0.31)	(0.30)
Kilometers from grid	19.06	15.75	19.89	−4.139***	−0.235	0.029
	(13.32)	(10.20)	(13.88)	(0.77)	(0.36)	(0.12)
Land gradient	10.10	9.12	10.35	−1.232***		
	(4.89)	(4.21)	(5.02)	(0.29)		
N communities	1,816	365	1,451	1,816	1,816	1,816

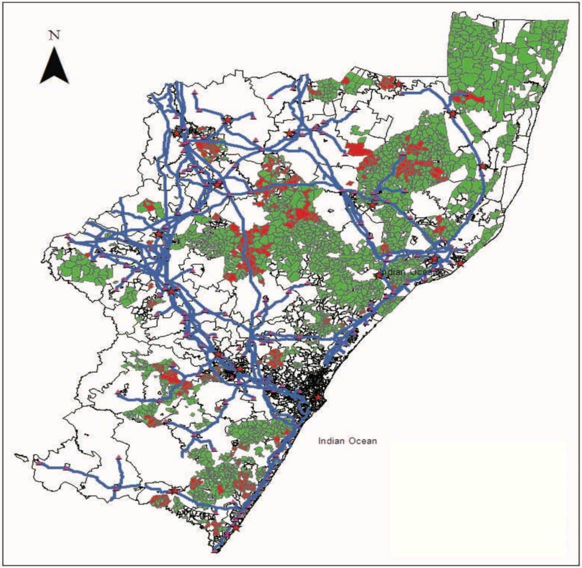
Dinkelman (2011): Data

	Year	Means (standard deviation) Full sample (1)	Eskom project (2)	No project (3)	Difference: Column 2–3 (4)
Female employment rate	1996	0.07	0.09	0.06	0.021***
		(0.08)	(0.07)	(0.08)	(0.00)
	2001	0.07	0.08	0.06	0.017***
		(0.07)	(0.07)	(0.07)	(0.00)
Difference	Δ_t	0.000	−0.003	0.001	−0.004
		(0.002)	(0.005)	(0.00)	(0.00)
Male employment rate	1996	0.14	0.16	0.13	0.031***
		(0.11)	(0.11)	(0.11)	(0.01)
	2001	0.10	0.11	0.10	0.014**
		(0.09)	(0.09)	(0.09)	(0.01)
Difference	Δ_t	−0.04***	−0.050***	−0.033***	−0.017***
		(0.00)	(0.01)	(0.00)	(0.01)
N		1,816	365	1,451	

Dinkelman (2011): Context



Dinkelman (2011): Data



Dinkelman (2011): Estimation approach

Electrification was not randomly assigned in South Africa:

- We need a research design to estimate the causal effect of interest

Ideally, we'd estimate:

$$Y_{id} = \alpha + \tau D_{id} + \varepsilon_{id}$$

where:

Y_{id} is the outcome (female employment rate) in community i in district d
D_{id} is an electrification indicator
ε_{id} is an error term

$\rightarrow$ Without random assignment, we will get bias (why?)

Dinkelman (2011): Estimation approach

Without random assignment, we could leverage time:

$$Y_{idt} = \tau D_{idt} + \alpha_{id} + \delta_t + \varepsilon_{idt}$$

$\rightarrow$ Note that Dinkelman writes this a bit weirdly

$$y_{jdt} = \alpha_0 + \alpha_1 t + \alpha_2 T_{jdt} + \mu_j + \delta_j t + \rho_d + \lambda_d t + \epsilon_{jdt},$$

Dinkelman (2011): Estimation approach

Without random assignment, we could leverage time:

$$Y_{idt} = \tau D_{idt} + \alpha_{id} + \delta_t + \varepsilon_{idt}$$

→ Note that Dinkelman writes this a bit weirdly

$$y_{jdt} = \alpha_0 + \alpha_1 t + \alpha_2 T_{jdt} + \mu_j + \delta_j t + \rho_d + \lambda_d t + \epsilon_{jdt},$$

- Why a time trend, not time FE?
- You can't have a district and a community FE?

Dinkelman (2011): Estimation approach

Without random assignment, we could leverage time:

$$Y_{idt} = \tau D_{idt} + \alpha_{id} + \delta_t + \varepsilon_{idt}$$

→ Note that Dinkelman writes this a bit weirdly

$$y_{jdt} = \alpha_0 + \alpha_1 t + \alpha_2 T_{jdt} + \mu_j + \delta_j t + \rho_d + \lambda_d t + \epsilon_{jdt},$$

- Why a time trend, not time FE?
- You can't have a district and a community FE?

→ Even with time, we still have identification concerns (why?)

Dinkelman (2011): Estimation approach

Dinkelman uses an IV approach to overcome the selection problem:

- We want to isolate the effect of electrification from everything else

For the instrument to be valid, we need:

1. **First stage:** Our IV needs to be correlated with electrification
2. **Exclusion restriction:** Our IV needs to only move Y through electrification

Dinkelman (2011): Estimation approach

Dinkelman uses an IV approach to overcome the selection problem:

- We want to isolate the effect of electrification from everything else

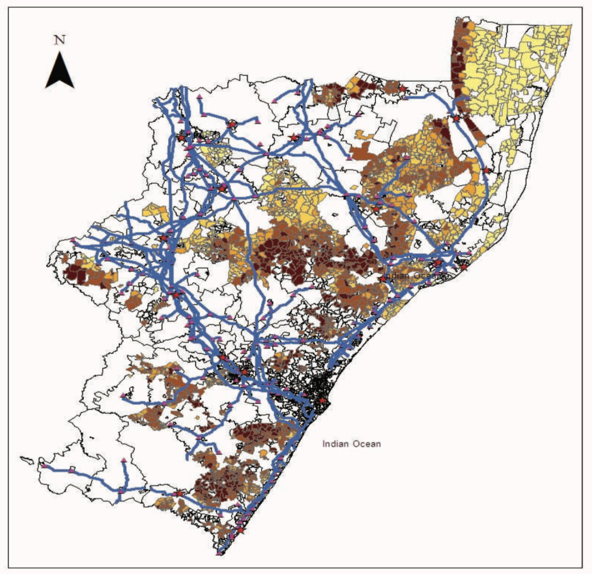
For the instrument to be valid, we need:

1. **First stage:** Our IV needs to be correlated with electrification
2. **Exclusion restriction:** Our IV needs to only move Y through electrification

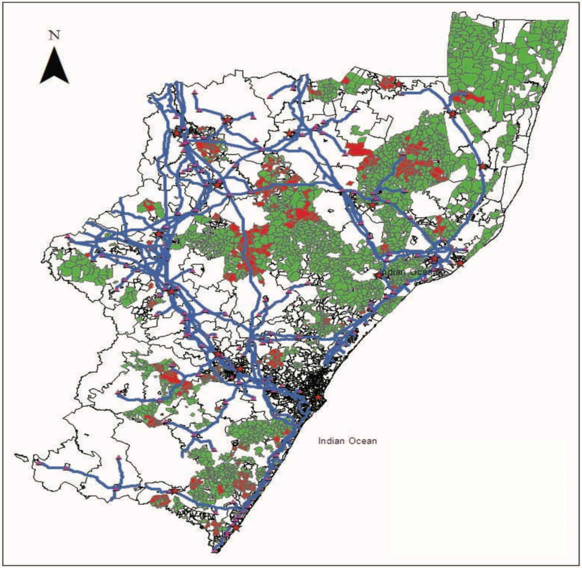
Instrument of choice: land gradient

- Steeper land is more expensive to electrify
- The first stage should be negative

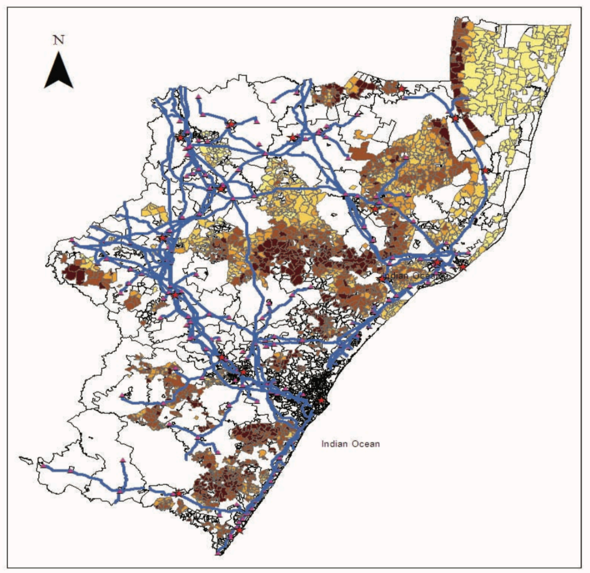
Dinkelman (2011): Estimation approach



Dinkelman (2011): Estimation approach



Dinkelman (2011): Estimation approach



Dinkelman (2011): Estimation approach

With the instrument, we simply estimate:

$$D_{idt} = \theta Z_{idt} + \alpha_{id} + \delta_t + \varepsilon_{idt}$$

$$Y_{idt} = \tau \hat{D}_{idt} + \alpha_{id} + \delta_t + \varepsilon_{idt}$$

Dinkelman (2011): Estimation approach

With the instrument, we simply estimate:

$$D_{idt} = \theta Z_{idt} + \alpha_{id} + \delta_t + \varepsilon_{idt}$$

$$Y_{idt} = \tau \hat{D}_{idt} + \alpha_{id} + \delta_t + \varepsilon_{idt}$$

Identifying assumption: Conditional on fixed effects, land gradient does not affect employment growth other than through electricity

- Is this reasonable?

Dinkelman (2011): Estimation approach

With the instrument, we simply estimate:

$$D_{idt} = \theta Z_{idt} + \alpha_{id} + \delta_t + \varepsilon_{idt}$$

$$Y_{idt} = \tau \hat{D}_{idt} + \alpha_{id} + \delta_t + \varepsilon_{idt}$$

Identifying assumption: Conditional on fixed effects, land gradient does not affect employment growth other than through electricity

- Is this reasonable?

LATEs: Estimates are the LATE for relatively flatter places

- How should this compare to ATE?

Dinkelman (2011): First stage

Dependent variable: Eskom project = [1 or 0]	(1)	(2)	(3)	(4)
Gradient × 10	−0.083**	−0.075**	−0.078***	−0.077***
	(0.040)	(0.034)	(0.027)	(0.027)
Kilometers to grid × 10		−0.040*	−0.012	−0.011
		(0.021)	(0.023)	(0.023)
Household density × 10		0.017***	0.012**	0.013**
		(0.004)	(0.006)	(0.006)
Poverty rate		0.023	0.019	0.017
		(0.069)	(0.070)	(0.069)
Female-headed HHs		0.393***	0.165	0.155
		(0.120)	(0.107)	(0.107)
Adult sex ratio		−0.173***	−0.130***	−0.121***
		(0.052)	(0.042)	(0.042)
Indian, white adults × 10		−1.236***	−1.116**	−1.105**
		(0.401)	(0.459)	(0.452)
Kilometers to road × 10		0.003	−0.010	−0.010
		(0.009)	(0.010)	(0.010)
Kilometers to town × 10		0.016	0.008	0.008
		(0.015)	(0.015)	(0.016)
Men with high school		−0.269	−0.185	−0.152
		(0.500)	(0.411)	(0.417)
Women with high school		1.046**	0.965**	0.984**
		(0.475)	(0.413)	(0.409)
Δ_t water access				0.012
				(0.048)
Δ_t toilet access				0.155
				(0.104)
District fixed effects	N	N	Y	Y
Mean of outcome variable	0.20	0.20	0.20	0.20
N communities	1,816	1,816	1,816	1,816
R^2	0.01	0.07	0.18	0.18
F-statistic on gradient	4.20	4.87	8.34	8.26
Pr > F	0.04	0.03	0.00	0.00

Dinkelman (2011): Household Behavior

Outcome is Δ_t in:	OLS No controls (1)	OLS Controls (2)	IV No controls (3)	IV Controls (4)
(1) Lighting with electricity	0.251***	0.221***	0.577***	0.635***
Mean: 0.08	(0.032)	(0.031)	(0.188)	(0.227)
(2) Cooking with wood	−0.045***	−0.039***	−0.266	−0.275*
Mean: −0.035	(0.012)	(0.012)	(0.179)	(0.147)
(3) Cooking with electricity	0.068***	0.056***	0.250**	0.228**
Mean: 0.037	(0.009)	(0.009)	(0.107)	(0.101)
(4) Water nearby	−0.029	0.005	−0.483*	−0.372
Mean: 0.007	(0.029)	(0.024)	(0.249)	(0.248)
(5) Flush toilet	0.003	0.008	0.018	0.067
Mean: 0.03	(0.006)	(0.005)	(0.069)	(0.068)

Dinkelman (2011): Female Employment

	Δ_t female employment rate							
	OLS regression coefficients				IV regression coefficients			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Eskom project	−0.004	−0.001	0.000	−0.001	0.025	0.074	0.090*	0.095*
	(0.005)	(0.005)	(0.005)	(0.005)	(0.045)	(0.060)	(0.055)	(0.055)
A. R. 95 percent C.I.							*[0.05; 0.3]*	*[0.05; 0.3]*
Poverty rate		0.029***	0.033***	0.031***		0.027**	0.032**	0.031**
		(0.011)	(0.010)	(0.010)		(0.012)	(0.013)	(0.013)
Female-headed HHs		0.042**	0.051***	0.047**		0.014	0.036	0.033
		(0.019)	(0.019)	(0.020)		(0.031)	(0.026)	(0.026)
Adult sex ratio		0.019**	0.017**	0.020***		0.033**	0.029**	0.032***
		(0.009)	(0.008)	(0.007)		(0.014)	(0.012)	(0.012)
Baseline controls?	N	Y	Y	Y	N	Y	Y	Y
District fixed effects?	N	N	Y	Y	N	N	Y	Y
Δ_t other services?	N	N	N	Y	N	N	N	Y
N communities	1,816	1,816	1,816	1,816	1,816	1,816	1,816	1,816

Dinkelman (2011): Male Employment

	Δ_t male employment rate							
	OLS regression coefficients				IV regression coefficients			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Eskom project	−0.017**	−0.015***	−0.009	−0.010*	−0.063	0.069	0.033	0.035
	(0.007)	(0.006)	(0.006)	(0.006)	(0.073)	(0.082)	(0.064)	(0.066)
A. R. 95 percent C.I.							*[−0.05; 0.25]*	*[−0.05; 0.25]*
Poverty rate		0.062***	0.064***	0.063***		0.059***	0.064***	0.062***
		(0.020)	(0.018)	(0.018)		(0.022)	(0.019)	(0.019)
Female-headed HHs		0.217***	0.233***	0.227***		0.187***	0.227***	0.220***
		(0.029)	(0.030)	(0.030)		(0.042)	(0.034)	(0.034)
Adult sex ratio		0.018*	0.012	0.017		0.034*	0.018	0.023
		(0.011)	(0.011)	(0.011)		(0.019)	(0.015)	(0.015)
Baseline controls?	N	Y	Y	Y	N	Y	Y	Y
District fixed effects?	N	N	Y	Y	N	N	Y	Y
Δ_t other services?	N	N	N	Y	N	N	N	Y
N communities	1,816	1,816	1,816	1,816	1,816	1,816	1,816	1,816

Dinkelman (2011): Panel Results

	Females		Males		Females		Males	
	OLS	FE	OLS	FE	OLS	FE	OLS	FE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Panel A. Employment [1/0]				*Panel B. Usual weekly hours of work*			
MD electrification rate	0.126**	0.128	0.090	0.134	6.646***	8.920	5.671**	13.090
	(0.058)	(0.149)	(0.077)	(0.164)	(1.771)	(6.634)	(2.597)	(12.947)
Trend (1995–2001)	−0.010	0.046**	−0.051***	−0.075***	−0.407	−0.588	−0.322	−1.424
	(0.012)	(0.020)	(0.012)	(0.022)	(0.491)	(0.872)	(0.620)	(1.701)
N	152	152	152	152	151	151	151	151
Mean of outcome	0.25	0.25	0.42	0.42	42.82	42.82	46.94	46.94
R^2	0.06	0.63	0.09	0.76	0.06	0.42	0.03	0.45
	Panel C. Log hourly wage				*Panel D. Log monthly earnings*			
MD electrification rate	−0.148	−1.380	0.101	0.171	−0.070	−0.616	0.414**	1.107**
	(0.253)	(1.046)	(0.211)	(0.483)	(0.225)	(0.995)	(0.191)	(0.477)
Trend (1995–2001)	−0.079***	0.132	−0.027	0.077	−0.091**	−0.065	−0.047	−0.085
	(0.030)	(0.137)	(0.032)	(0.063)	(0.037)	(0.131)	(0.033)	(0.063)
N	146	146	148	148	146	146	148	148
Mean of outcome	1.17	1.17	1.49	1.49	6.42	6.42	6.80	6.80
R^2	0.03	0.52	0.00	0.51	0.03	0.52	0.05	0.57

A slight digression: multiple testing

The "reproducibility crisis" is becoming a Thing:

- We have lots of results that don't replicate
- When we try to re-do experiments, we don't find the same results
- Famous example: power poses!

A slight digression: multiple testing

The "reproducibility crisis" is becoming a Thing:

- We have lots of results that don't replicate
- When we try to re-do experiments, we don't find the same results
- Famous example: power poses!

→ A central culprit: **p-hacking**

  - Researchers get jobs based on statistically significant effects
  - This generates incentives to find them
  - We get a lot of results with $0.051 < p < 0.043$
  - ... a lot more than 20%!
  - Adding controls, etc to get "stars" is common

A slight digression: multiple testing

One way to generate "stars" is to run a lot of regressions:

- With a 95% confidence threshold, you will find stars 5% of the time, even if your null hypothesis is true
- So if you run a lot of regressions, you are bound to find some with stars
- If you then only report the ones with stars, we have a problem
- (If you report all of them, we're fine!)

A slight digression: multiple testing

One way to generate "stars" is to run a lot of regressions:

- With a 95% confidence threshold, you will find stars 5% of the time, even if your null hypothesis is true
- So if you run a lot of regressions, you are bound to find some with stars
- If you then only report the ones with stars, we have a problem
- (If you report all of them, we're fine!)

→ If we do this, we're killing the usefulness of stars

→ And generating results that won't replicate

Addressing multiple testing

There are a few fixes to this issue:

1. Pre-specification
   - Before you look at data, write down exactly what you're going to run
   - And make this public
   - So I can tell how many tests you actually did!

   $\rightarrow$ To make this credible, need to prove you couldn't see data first

Addressing multiple testing

There are a few fixes to this issue:

1. Pre-specification
   - Before you look at data, write down exactly what you're going to run
   - And make this public
   - So I can tell how many tests you actually did!

   → To make this credible, need to prove you couldn't see data first

2. Multiple correction adjustment
   - Adjust for the fact that you ran many tests
   - Essentially involves inflating your p-values
   - Many ways to do this: FWER, FDR, Bonferroni

   → This is best paired with prespecification

Dinkelman (2011): Spillovers?

Outcome: Δ_t female employment	OLS (1)	IV (2)	*N* communities (3)
Panel A.			
Full sample	−0.001	0.095*	1,816
	(0.005)	(0.055)	
Panel B.			
Excluding nonproject areas $<$ 1 km from project site	−0.004	0.076	1,205
	(0.006)	(0.057)	
Panel C.			
Excluding nonproject areas $<$ 5 km from project site	−0.003	0.069	840
	(0.008)	(0.077)	

Dinkelman (2011): Migration

	Δ_t log population		Δ_t females with high school		Δ_t males with high school	
	OLS	IV	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A.						
Eskom project	0.171***	3.897***	0.001	0.129*	0.001	0.076
	(0.045)	(1.427)	(0.005)	(0.058)	(0.003)	(0.050)
N	1,816	1,816	1,816	1,816	1,816	1,816
	Δ_t log non–in-migrant population		Δ_t female employment excluding in-migrants		Δ_t male employment excluding in-migrants	
Panel B.						
Eskom project	0.181***	4.349***	0.000	0.116*	−0.008	0.086
	(0.048)	(1.586)	(0.005)	(0.069)	(0.005)	(0.069)
N	1,816	1,816	1,816	1,816	1,816	1,816

Recap

TL;DR:

1. Dinkelman (2011) is a seminal study of the effects of rural electrification
2. Finds that electrification dramatically increases female employment
3. Uses an IV strategy based on land gradient (credible?)

For next class

Topics:

- Policy lab: Does rural electrification work? II

Readings:

- Lee, Miguel, and Wolfram (2019)
- Burlig and Preonas (2016)